Name: _____ Mark: _____ / 9

Mini-math Div 3/4: Wednesday, October 21, 2026 (6.1-6.8) - 18 minutes

1. (1 point) Water flows into a tank at a rate of $r(t)$ liters per minute, where t is in minutes. Which expression represents the total amount of water that flows into the tank from $t = 2$ to $t = 5$ minutes?

   A. $r(5) - r(2)$ B. $r'(5) - r'(2)$ C. $\int_2^5 r(t)\ dt$ D. $\int_2^5 r'(t)\ dt$ E. $r(5) \cdot (5 - 2)$

2. (1 point) Suppose $V(x) = \int_0^{x^2} \sin t\, dt$. What is the derivative, $V'(x)$?

   A. $\cos x$ B. $\sin x$ C. $\sin x^2$ D. $2x \sin x^2$ E. $2x \cos x^2$

3. (1 point) $\int_1^8 t^{-2/3}\, dt =$

   A. -3 B. -1 C. $\frac{93}{160}$ D. 1 E. 3

4. (1 point) Given $\int_1^7 f(x)\,dx = 4$, $\int_{-1}^7 f(x)\,dx = -3$, and $\int_1^5 f(x)\,dx = 6$, find $\int_{-1}^5 (2f(x)+3)\,dx$

A. -2 B. 15 C. 16 D. 17 E. 28

5. (1 point) Evaluate $\int_0^{\pi/2} (2\cos x - \sin x)\,dx$.

A. 1 B. 2 C. 3 D. $\pi - 1$ E. $\pi + 1$

6. (1 point) Choose the integral that is the limit of the Riemann Sum: $\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(1+\frac{8k}{n}\right)\cdot\frac{4}{n}$

A. $\int_0^4 \sin(1+2x)\,dx$ B. $\int_1^5 \sin(1+x)\,dx$ C. $\int_1^5 \sin(1+2x)\,dx$ D. $\int_1^5 \sin x\,dx$

7. (1 point) Suppose f is concave up and the following are selected values of f:

x	0	1	3	4	6
$f(x)$	3	2	4	6	12

Using the trapezoidal rule with 4 unequal subintervals to approximate $\int_0^6 f(x)\,dx$, then:

A. $\int_0^6 f(x)\,dx \approx 31.5$ and this is an underestimate

B. $\int_0^6 f(x)\,dx \approx 31.5$ and this is an overestimate

C. $\int_0^6 f(x)\,dx \approx 63$ and this is an underestimate

D. $\int_0^6 f(x)\,dx \approx 63$ and this is an overestimate

8. (1 point) Choose the limit of the Riemann Sum that is the integral: $\displaystyle\int_2^4 \frac{1}{x+2}\,dx$

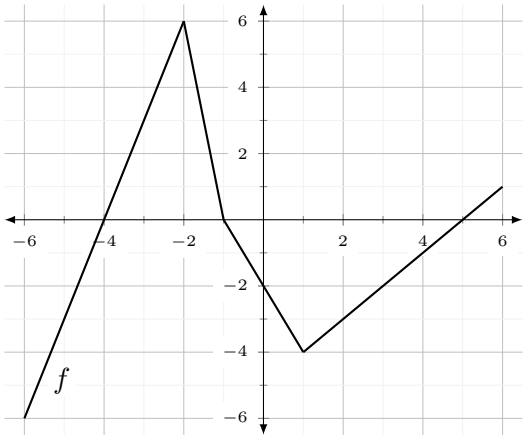
A. $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{\frac{k}{n}+2}\cdot\left(\frac{2}{n}\right)$

B. $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{\frac{2k}{n}+2}\cdot\left(\frac{2}{n}\right)$

C. $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{\frac{k}{n}+4}\cdot\left(\frac{2}{n}\right)$

D. $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{\frac{2k}{n}+4}\cdot\left(\frac{2}{n}\right)$

9. (1 point) To the right is a graph of $f(x)$. Suppose $g(x)=\displaystyle\int_a^x f(t)\,dt$ and $g(1)=3$. What is the minimum value of $g(x)$ on $[-6,2]$?



A. -8

B. -5

C. -4

D. -3

E. -2